

Retail Customer Intelligence & Loyalty Optimization

End-to-End Business Analytics Case Study

Python | SQL | Power BI

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1. Executive Summary

The business generated **\$233K in revenue** from **3.9K customers**, with an **average spend of \$59.76** and a **customer satisfaction score of 3.75/5**, indicating stable overall performance.

Revenue was primarily driven by **Clothing (\$104.3K)** and **Accessories (\$74.2K)**, while **Outerwear (\$18.5K)** underperformed, highlighting category-level optimization opportunities.

The business maintained a strong **79% loyal customer base**, while retention efficiency remained high at **97.87%**, indicating strong repeat purchase continuity.

Subscribers contributed only **27% of revenue**, while their average spend (**\$59.49**) remained nearly identical to non-subscribers (**\$59.86**), indicating that the current subscription model is not creating incremental spending value.

Overall, the analysis indicates a stable revenue base with opportunities to improve performance through **subscription growth, targeted customer engagement, and category optimization**.

2. Problem Statement and Business Objectives

A leading retail company is experiencing inconsistent revenue growth and declining repeat purchase behavior across customer segments, product categories, and sales channels.

The objective of this project is to identify the key drivers of customer spending, loyalty behavior, category performance, and retention using demographic patterns, seasonal demand, customer reviews, promotional strategies, and purchase history.

Business Objectives

- Identify high-value customer segments
- Improve customer retention strategies
- Evaluate subscription and loyalty performance
- Analyze product category profitability
- Optimize shipping and promotional strategies
- Develop executive-level dashboards for business decision-making

3. Dataset Overview and Project Workflow

The project followed an end-to-end analytics workflow starting from raw retail customer transaction data and progressing through data preprocessing, SQL-based business analysis, and Power BI dashboard development to generate business insights related to customer behavior, loyalty, and revenue performance.

3.1 Dataset Overview

The analysis was performed using a retail customer shopping behavior dataset containing demographic, transactional, and customer loyalty-related information.

- Rows: 3900
- Columns after preprocessing: 18
- Engineered columns:
 - age_group
 - purchase_frequency_days
- Domain - Retail Customer Analytics

3.2 Key Features

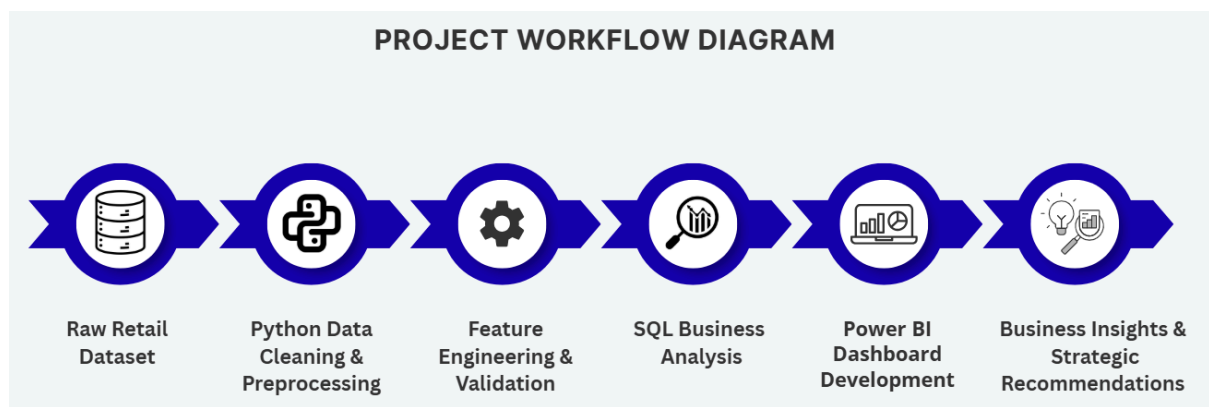
- customer_id
- age
- gender
- category
- purchase_amount
- review_rating
- subscription_status
- shipping_type
- previous_purchases
- frequency_of_purchases

3.3 Engineered Features

- age_group
- purchase_frequency_days

3.4 Project Workflow

The project workflow included data preprocessing in Python, business analysis in SQL, and dashboard development in Power BI.



4. Python Data Cleaning and Preprocessing

Data cleaning and preprocessing were performed in Python using Google Colab to improve dataset consistency and prepare the data for SQL analysis and Power BI dashboard development.

4.1 Preprocessing Workflow

The preprocessing process included:

- Dataset inspection
- Missing value handling
- Data validation
- Column standardization
- Feature engineering
- Clean dataset export

df.head()

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express	Yes	Yes	14
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express	Yes	Yes	2
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Free Shipping	Yes	Yes	23
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	Next Day Air	Yes	Yes	49
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Free Shipping	Yes	Yes	31

4.2 Missing Value Handling

Missing values in the `review_rating` column were handled using category-wise median imputation to maintain consistency in customer review analysis while reducing the impact of outliers.

```
# Handling Missing Values
df['Review Rating'] = df.groupby('Category')['Review Rating'].transform(lambda x: x.fillna(x.median()))
```

4.3 Data Validation

Validation checks were performed to identify inconsistencies between `discount_applied` and `promo_code_used`. After validation, the `promo_code_used` column was removed to eliminate redundancy.

4.4 Column Standardization

Column names were converted into lowercase and underscore-separated format to improve readability and maintain consistency across Python, SQL, and Power BI workflows.

Before		After
Purchase Amount (USD)	:	purchase_amount
Review Rating	:	review_rating

4.5 Feature Engineering

Additional features were created to support customer segmentation and behavioral analysis.

Feature	Purpose
Age_group	Customer segmentation by age category
Purchase_frequency_days	Numerical representation of purchase behavior

The `age_group` feature was created using quantile-based segmentation, while `purchase_frequency_days` converted categorical purchase frequency into a measurable analytical metric.

```
[16]
✓ Os
# Creating Age Groups

import pandas as pd
labels = ['Young Adult', 'Adult', 'Middle-aged', 'Senior']
df['age_group'] = pd.qcut(df['age'], q=4, labels=labels)
```

4.6 Final Dataset Preparation

After preprocessing and validation, the cleaned dataset was exported for SQL analysis and Power BI dashboard development. Python preprocessing was performed using Pandas in Google Colab to improve dataset quality and prepare the data for business analysis.

```
[28]
✓ Os
# Final Cleaned Dataset

df.head()
```

	customer_id	age	gender	item_purchased	category	purchase_amount	location	size	color	season	review_rating	subscription
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	

5. SQL Business Analysis

After preprocessing the retail dataset in Python, SQL was used to analyze customer purchasing behavior, category performance, loyalty trends, and promotional effectiveness within the retail customer dataset. The objective of this phase was to identify patterns influencing customer spending, repeat purchasing behavior, and revenue performance to support business-focused decision-making and dashboard development.

The cleaned dataset was imported into the `retail_project` database and analyzed using aggregation functions, CASE statements, Common Table Expressions (CTEs), ranking functions, and segmentation logic.

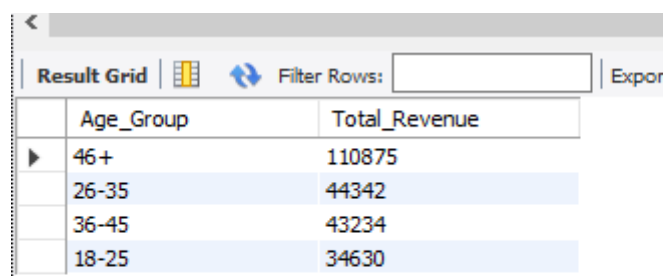
5.1 Customer Behavior and Loyalty Analysis

SQL analysis was performed to understand customer purchasing behavior across different customer segments, subscription groups, and purchase frequency patterns. Customer lifecycle segmentation was also conducted to classify customers into New, Returning, and Loyal groups based on purchase history.

Key Insights

- Customers were segmented into **New, Returning, and Loyal** groups to understand repeat purchasing behavior and engagement patterns.
- Subscriber and non-subscriber spending analysis identified differences in purchasing behavior, with total spending values of **\$62,645** and **\$170,436**, respectively.
- Customer age-segment analysis showed that the **46+** age segment contributed the highest purchase amount of **\$110,875**.
- The loyalty model identified a **small but valuable loyal segment (57 customers)**, while majority customers showed lower loyalty potential, indicating strong opportunity for retention programs.

The analysis helped identify customer groups contributing more actively to purchasing activity and highlighted how subscription behavior, purchase history, and customer demographics influence business performance.



	Age_Group	Total_Revenue
▶	46+	110875
	26-35	44342
	36-45	43234
	18-25	34630

5.2 Revenue and Product Performance Analysis

Revenue-focused SQL analysis was conducted to identify top-performing categories and products contributing to sales performance. Category-level revenue comparisons and ranking functions were used to evaluate product performance.

Key Insights

- **Clothing** recorded the highest category revenue of **\$104,264**, while **Outerwear** showed comparatively lower performance.
- Product ranking analysis identified the top revenue-generating products within each category using **DENSE_RANK()**.
- Customers within the **Average(3.00 - 3.99)** segment recorded the highest purchase frequency of **1,586** and generated total revenue of **\$94,400**. Purchase frequency remains relatively stable across rating segments, with only marginal variation.
- **Sneakers in Footwear Category** recorded lower sales performance with revenue of **\$8,635** indicating potential need for promotional support and inventory optimization.

The analysis highlighted revenue concentration across categories and products, helping identify areas for inventory prioritization and performance optimization.

	Category	Item_Purchased	Demand_Frequency	Revenue	Demand_Rank
▶	Accessories	Jewelry	171	10010	1
	Accessories	Sunglasses	161	9649	2
	Accessories	Belt	161	9635	2
	Clothing	Blouse	171	10410	1
	Clothing	Pants	171	10090	1
	Clothing	Shirt	169	10332	3
	Footwear	Sandals	160	9200	1
	Footwear	Shoes	150	9240	2
	Footwear	Sneakers	145	8635	3
	Outerwear	Jacket	163	9249	1
	Outerwear	Coat	161	9275	2

5.3 Retention and Promotional Analysis

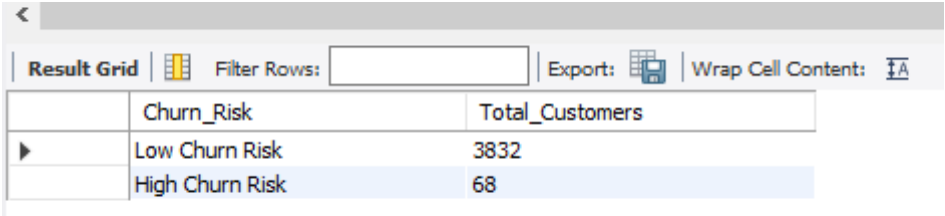
Additional SQL analysis was performed to evaluate discount effectiveness, identify churn-risk customers, and determine customer groups suitable for targeted promotional campaigns.

Key Insights

- Discount-based purchases generated total revenue of **\$99,411**, compared to **\$133,670** from non-discounted purchases, indicating the influence of promotional offers on customer spending behavior.
- Overall churn risk remains low, with only **68 customers classified as high churn risk**, but this segment requires proactive intervention due to revenue risk.

- Revenue ranking analysis identified **Outerwear** as a lower-performing segment with revenue contribution of **\$18,524**, indicating the need for stronger promotional focus and inventory optimization.

The analysis revealed that discounts and promotional strategies positively influenced purchasing activity, while churn-risk segmentation helped identify customer groups requiring stronger retention and engagement efforts to reduce potential revenue loss.



The screenshot shows a Power BI interface with a table titled 'Result Grid'. The table has two columns: 'Churn_Risk' and 'Total_Customers'. The data is as follows:

Churn_Risk	Total_Customers
Low Churn Risk	3832
High Churn Risk	68

Key SQL Concepts Applied

- Aggregate Functions
- CASE Statements
- Common Table Expressions (CTEs)
- Window Functions (**RANK()** and **DENSE_RANK()**)
- Customer Segmentation Logic
- Conditional Analysis

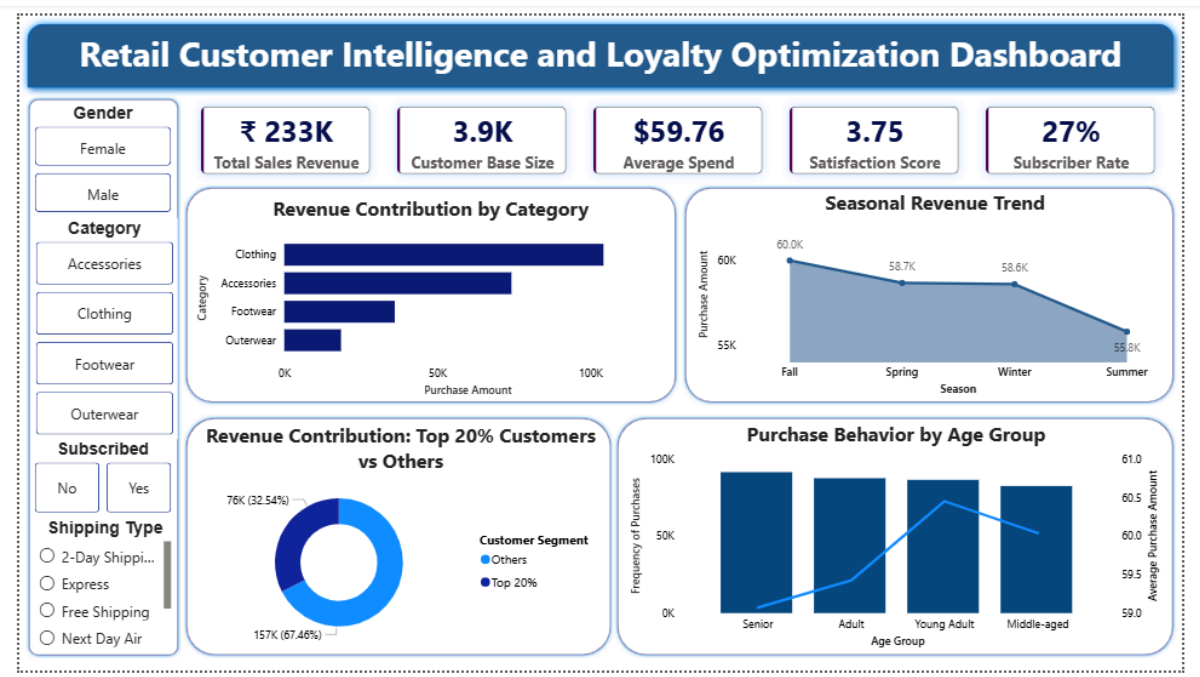
The insights generated through SQL analysis supported the Power BI phase of this Project by contributing to KPI selection, customer behavior analysis, revenue performance tracking, and business recommendation development.

6. Dashboard 1: Retail Customer Intelligence and Loyalty Optimization Dashboard

This dashboard evaluates overall retail performance by analyzing revenue distribution across product categories, customer segments, seasonal demand patterns, and purchasing behavior to identify the primary drivers of revenue concentration and customer value.

6.1 Performance Snapshot

The business generated **\$233K in total revenue** from a customer base of **3.9K customers**, with an average customer spend of **\$59.76** and a satisfaction score of **3.75**. Subscriber penetration remains limited at **27%**, indicating that the majority of revenue is currently driven by non-subscribed customers.



6.2 Revenue Distribution Across Product Categories

Category analysis revealed a clear revenue concentration in **Clothing (\$104.3K)**, making it the strongest revenue contributor, followed by **Accessories (\$74.2K)**. In contrast, **Outerwear (\$18.5K)** and **Footwear (\$36.1K)** delivered comparatively lower contributions, highlighting underperforming product segments and opportunities for targeted promotional and inventory optimization strategies.

The revenue gap between the top-performing and lower-performing categories highlights an uneven category contribution structure, where overall sales performance depends heavily on a limited product mix.

6.3 Seasonal Revenue Movement

Seasonal demand remains relatively stable throughout Fall, Spring, and Winter, with Fall recording the highest revenue contribution at nearly **\$60K**. Revenue declines during Summer to approximately **\$55K**, representing the weakest-performing seasonal period.

Seasonal revenue remained relatively stable within a narrow range of approximately **\$55K–\$60K**, indicating balanced year-round demand with only minor seasonal fluctuations.

6.4 Revenue Concentration Among High-Value Customers

Revenue generation is highly concentrated within a relatively small customer segment. The top 20% of customers contribute **\$157K (67.46%)** of total sales, whereas the remaining customer base contributes only **\$76K (32.54%)**.

This distribution demonstrates a clear high-value customer dependency, where a limited segment accounts for more than two-thirds of overall business revenue.

6.5 Purchase Behavior Across Age Groups

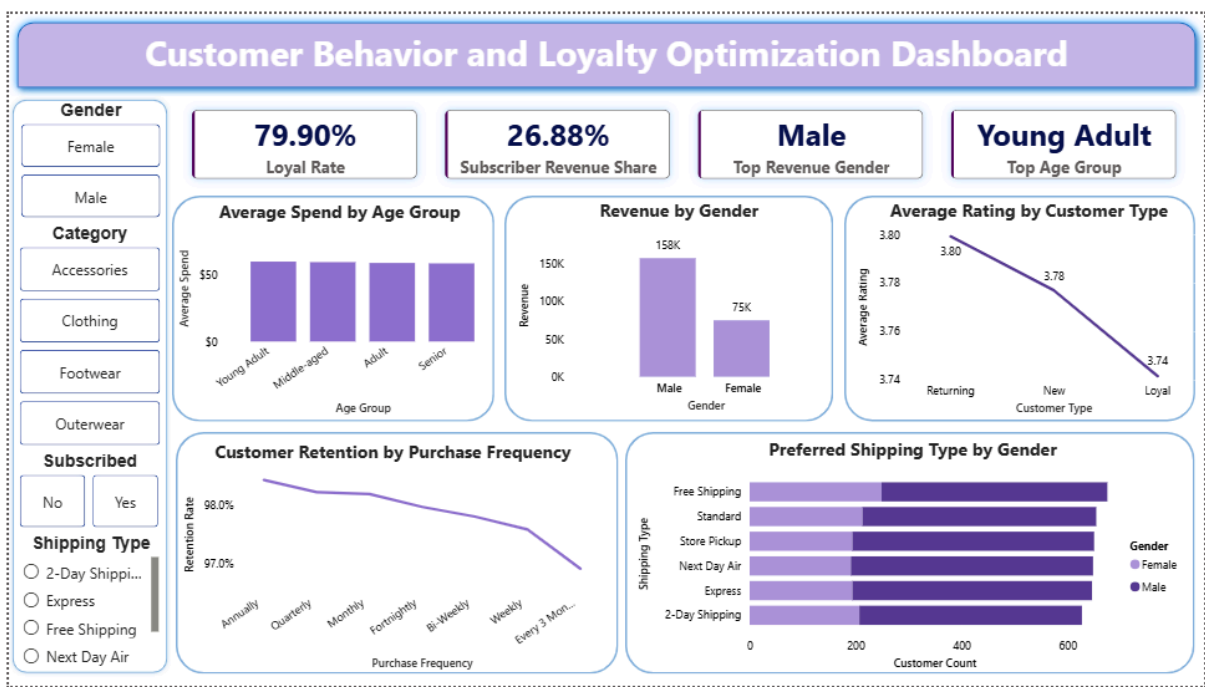
Senior customers showed the **highest cumulative purchase frequency score**, indicating stronger repeat buying behavior relative to other age groups. In contrast, Young Adults generate the highest average purchase value at approximately **\$60.5 per transaction**.

The dashboard therefore reveals two distinct behavioral patterns:

- older customers contribute through purchasing consistency
- younger customers contribute through higher spending intensity

7. Dashboard 2: Customer Behavior and Loyalty Optimization Dashboard

This dashboard analyzes customer loyalty, demographic contribution, spending behavior, satisfaction trends, retention patterns, and shipping preferences to evaluate the behavioral factors influencing long-term revenue performance and customer retention.



7.1 Performance Snapshot

The business maintained a strong **79% loyalty rate**, indicating healthy repeat purchase behavior across the customer base. However, **subscriber revenue share remained limited at 27%**, showing that a significant portion of revenue continues to come from non-subscribed customers. Revenue contribution was also concentrated among specific demographic groups, with **male customers generating \$158K** and the **46+ age segment contributing \$110.9K**, highlighting dependence on a limited set of high-value customer segments.

7.2 Gender Revenue Analysis

Revenue contribution showed a significant gender imbalance, with **male customers contributing \$158K**, more than double the **\$75K generated by female customers**. This indicates that current sales performance is heavily dependent on one demographic segment, creating an opportunity to improve revenue diversification through more targeted customer engagement strategies.

7.3 Average Spend by Age Group

While older customers contributed a larger share of total revenue overall, **Young Adults recorded the highest average spend per transaction**, making them the strongest segment in terms of average order value. This suggests that younger customers represent a high-value segment with stronger spending efficiency despite lower overall revenue contribution.

7.4 Customer Satisfaction Analysis

Customer satisfaction showed a gradual decline across the customer lifecycle, with ratings decreasing from **3.80 among Returning customers** to **3.74 among Loyal customers**. Although the decline is moderate, it suggests minor opportunities to improve long-term experience consistency.

7.5 Retention Trend Analysis

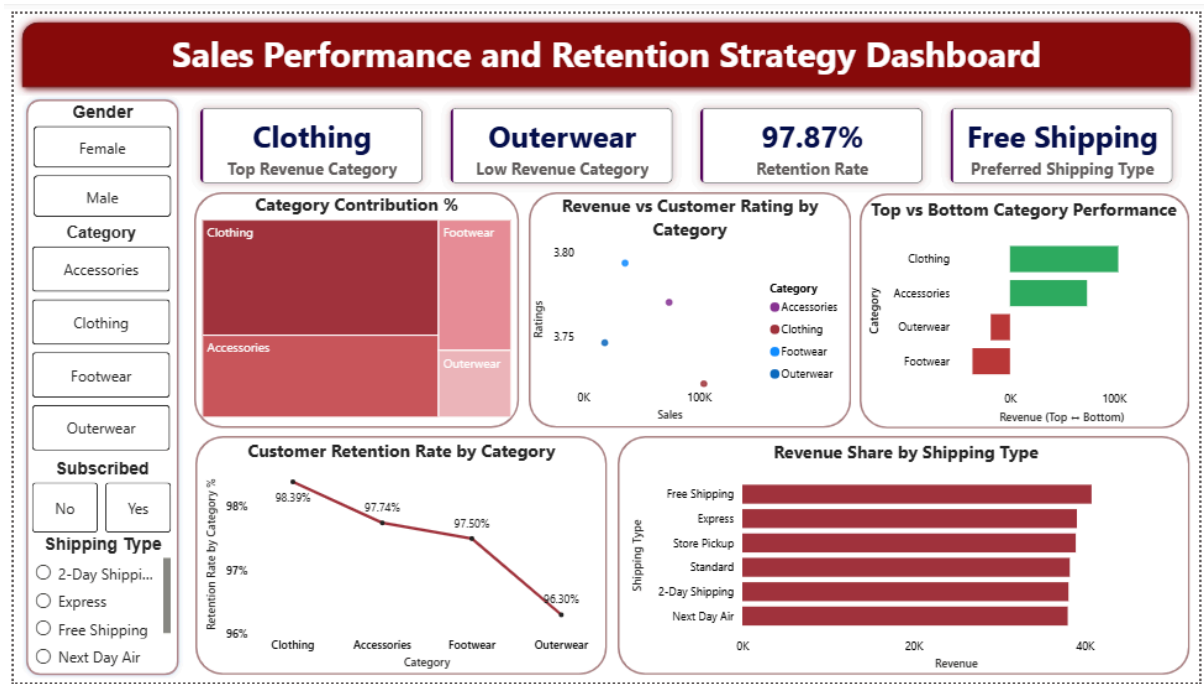
Customer retention remained strongest among **Annual and Quarterly purchasers**, while lower-frequency buyers showed comparatively weaker retention behavior. This highlights a positive relationship between purchasing consistency and long-term customer loyalty, suggesting that encouraging repeat purchasing behavior can strengthen retention performance.

7.6 Shipping Preference Analysis

Free Shipping and Standard Shipping emerged as the most preferred fulfillment options across customer groups, while premium delivery methods showed lower adoption. This indicates that customer shipping decisions are influenced more by cost efficiency than faster delivery speed, making affordability a stronger driver of purchase convenience.

8. Dashboard 3: Sales Performance and Retention Strategy Dashboard

This dashboard evaluates category-level sales contribution, customer retention patterns, product performance, customer satisfaction impact, and shipping-based revenue distribution to identify the key operational drivers influencing revenue growth and retention efficiency.



8.1 Performance Snapshot

Clothing emerged as the strongest revenue category, generating **\$104K** and contributing the largest share of total category sales. **Outerwear** remained the weakest segment with only **\$18.5K** in revenue, highlighting underperformance in product contribution. The business also maintained an exceptionally high **97.87% retention rate**, indicating strong customer continuity, while **Free Shipping** emerged as the most revenue-generating shipping preference.

8.2 Category Contribution Analysis

Revenue contribution was heavily concentrated in a few product categories, with **Clothing contributing the largest share**, followed by **Accessories**, while **Footwear and Outerwear contributed significantly lower portions** of total sales. This indicates a revenue concentration pattern where business performance is dependent on a limited number of high-performing categories.

8.3 Revenue vs Customer Rating by Category

Category-level comparison showed that **Clothing generated the highest revenue despite not having the highest customer rating**, while **Footwear recorded the strongest customer rating with comparatively lower sales contribution**. This suggests that strong customer satisfaction alone does not always translate into higher revenue, indicating opportunities for better monetization of well-rated but lower-performing categories.

8.4 Top vs Bottom Category Performance

A clear performance gap exists between top and bottom product categories, with **Clothing and Accessories significantly outperforming Outerwear and Footwear in revenue generation**. The sharp difference highlights category imbalance and suggests that underperforming segments may require stronger promotional support, pricing review, or assortment optimization.

8.5 Customer Retention Rate by Category

Retention performance remained consistently strong across most categories, with **Clothing recording the highest retention at 98.39%**, followed by **Accessories and Footwear**, while **Outerwear declined to 96.30%**, making it the weakest category in retention performance as well. This suggests that lower sales performance in Outerwear is also accompanied by weaker customer stickiness.

8.6 Revenue Share by Shipping Type

Free Shipping generated the highest revenue contribution, followed closely by **Express and Store Pickup**, while premium fast-delivery options contributed slightly lower revenue shares. The relatively close distribution suggests that shipping convenience influences revenue, but affordability remains the stronger purchase driver.

9. Strategic Insights and Business Recommendations

This section summarizes the key business insights derived from customer segmentation, revenue analysis, churn prediction, subscription behavior, and retention analytics, and provides strategic recommendations to improve customer value, profitability, and long-term business growth for this retail business.

9.1 Key Strategic Insights

1. Revenue Concentration Creates Dependency on a Small Customer Segment

The business generates **67.46% of its total revenue (\$157K)** from the top 20% of customers, showing that a relatively small group contributes a disproportionately large share of business performance. While these customers are highly valuable, this level of concentration creates dependency risk because any drop in retention or spending within this segment could significantly affect overall revenue stability. Strengthening value contribution from a broader customer base would create a more balanced and resilient revenue structure.

2. Category Performance Is Uneven Across the Product Portfolio

Product revenue is concentrated in a few categories, with **Clothing generating \$104,264** and **Accessories contributing \$74,200**, while **Outerwear generated only \$18,524**. This shows that current revenue performance is being driven by a limited product mix, while some categories remain underutilized. A more balanced category portfolio would help reduce revenue dependency on a few strong-performing segments and support broader growth opportunities.

3. Customer Retention Is Strong, but New Customer Acquisition Remains Limited

The business demonstrates strong customer loyalty, with **79.9% of customers (3,116 customers)** classified in the loyal segment. However, the business has only **83 new customers**, indicating that current growth is being driven more by retention than customer acquisition. While strong retention supports revenue stability, limited acquisition may constrain long-term growth if the customer base does not continue to expand.

4. Subscription Membership Is Not Delivering Higher Spending Value

Subscribers recorded an average spend of **\$59.49**, compared to **\$59.86 for non-subscribers**, showing almost no meaningful difference in customer spending behavior. This suggests that while the subscription model may support customer participation, it is not currently creating additional monetization value. The subscription program therefore appears to have engagement value but limited commercial impact in its current form.

5. Revenue Contribution Is Concentrated Within Specific Customer Segments

Revenue contribution is also concentrated within specific demographic groups, with **male customers generating approximately \$158K**, compared to **\$75K from female customers**, while the **46+ age segment contributed \$110,875**, making it the highest-value age group. This reflects dependence on a limited set of customer segments and highlights the need to diversify revenue contribution across broader demographic groups.

6. Strong Retention Performance Still Requires Revenue Protection

The business maintained a strong **97.87% retention rate**, with only **68 customers identified as high churn risk**. While overall retention remains a business strength, even a small churn-prone segment represents preventable revenue leakage if not addressed proactively. Protecting these customers can help preserve recurring revenue and strengthen long-term customer value.

7. Broad Discounting Is Not Creating Meaningful Spending Uplift

Customers purchasing without discounts recorded a slightly higher average spend of **\$60.13**, compared to **\$59.28 for discount-driven purchases**. This suggests that discounting may be encouraging purchases, but it is not significantly increasing customer spending value. Over time, broad discounting without clear spending uplift may weaken profitability and reduce promotional efficiency.

8. Seasonal Revenue Remains Relatively Stable Across the Year

Seasonal revenue remained within a narrow range of **\$55K–\$60K**, indicating relatively stable demand throughout the year with limited dependence on major seasonal spikes. This reflects a balanced baseline revenue structure supported by consistent purchasing behavior across seasonal cycles.

9.2 Strategic Recommendations

Based on the analytical findings, the following strategic recommendations are proposed:

- **Strengthen premium loyalty programs** for high-value customers to reduce revenue concentration risk and improve customer lifetime value.
 - **Expand and prioritize high-performing categories**, especially Clothing, while reviewing pricing, assortment, and promotional strategies for underperforming categories such as Outerwear.
 - **Increase customer acquisition initiatives**, particularly targeting younger customers and underrepresented demographic segments to improve long-term market diversification.
 - **Redesign the subscription program** by introducing exclusive rewards, premium benefits, and spending-based incentives to improve monetization effectiveness.
 - **Deploy targeted churn prevention strategies** for high-risk customers through personalized offers, engagement campaigns, and retention-focused interventions.
 - **Replace blanket discounting with data-driven targeted promotions** to improve campaign efficiency while protecting profitability.
 - **Use customer satisfaction insights as an early warning indicator** to identify dissatisfaction and trigger proactive customer retention actions.
-

10. Expected Business Impact

1. Improved Revenue Stability Through Reduced Concentration Risk

Reducing dependence on the top 20% of customers, who currently contribute **67.46% of total revenue (\$157K)**, would create a more balanced customer revenue mix and reduce business risk associated with overreliance on a small high-value segment.

2. Higher Customer Lifetime Value Through Loyalty and Subscription Optimization

Improving loyalty engagement for the existing **3,116 loyal customers** and redesigning the subscription model for **1,053 subscribers** can help strengthen repeat purchasing behavior, improve monetization effectiveness, and increase long-term customer value.

3. Sustainable Revenue Growth Through Customer Acquisition Expansion

Expanding beyond the current **83 new customers** would help diversify the customer base, reduce dependence on existing loyal customers, and support stronger long-term revenue growth opportunities.

4. Improved Profitability Through Smarter Promotional Strategy

Since non-discount customers already spend slightly more (**\$60.13 vs \$59.28**), replacing broad discounting with targeted promotions can improve promotional efficiency, protect margins, and generate better return on marketing investment.

5. More Balanced Category-Level Revenue Contribution

Improving lower-performing categories such as **Outerwear (\$18,524 revenue)** while strengthening high-performing categories like **Clothing (\$104,264 revenue)** can create a more balanced product portfolio and reduce category-level revenue dependency.

6. Stronger Revenue Protection Through Churn Prevention

Targeted retention interventions for the **68 high-risk customers** can help protect recurring revenue, minimize preventable customer loss, and support the business's strong **97.87% retention performance**.

7. Greater Market Diversification and Segment Expansion

Increasing engagement among underrepresented customer groups, particularly female customers and younger age segments, can help reduce demographic revenue concentration and support broader long-term market expansion.

11. Conclusion

This retail analytics project demonstrates that the business operates from a position of **strong operational stability**, generating approximately **\$233K in total revenue**, maintaining a strong repeat-purchase continuity indicator of **97.87%**, and benefiting from a highly loyal customer base where **79.9% of customers are repeat purchasers**. These findings reflect effective customer retention and a stable revenue foundation.

However, the analysis also reveals important strategic risks that may affect long-term growth, including **67.46% revenue concentration within the top 20% customers**, limited first-time purchase representation in the observed dataset (**83 customers**), category-level performance imbalance, demographic revenue concentration, and limited subscription monetization effectiveness.

To achieve sustainable long-term business growth, the company must move beyond retention-led performance and focus on **customer acquisition, revenue diversification, category optimization, loyalty redesign, and targeted profitability improvement**. By implementing these strategic priorities, the business can strengthen profitability, reduce structural business risk, improve customer lifetime value, and build a more resilient and competitive retail growth model.
